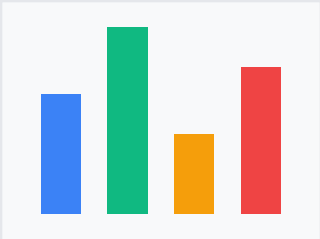
Comprehensive Data Visualization Reference Manual



1. Bar Chart

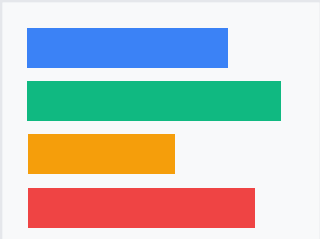
Purpose: Compares categorical data sizes cleanly using vertical rectangular blocks.

Syntax:
```
plt.bar(categories, values)
```

2. Scatter Plot

Purpose: Tracks coordinates across independent and dependent vectors to assess statistical correlation.

Syntax:
```
plt.scatter(x, y)
```



3. Horizontal Bar Chart

Purpose: Compares categorical metrics efficiently when text descriptors are exceptionally long.

Syntax:
```
plt.barh(categories, values)
```

4. Line Chart

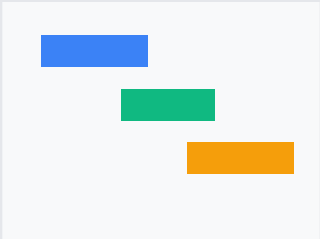
Purpose: Tracks steady progressive movements, continuous cycles, and directional momentum trends over uniform scales.

Syntax:
```
plt.plot(timeline, values)
```

5. Area Chart

Purpose: Emphasizes the raw volumetric depth underneath progress tracking lines over chronological durations.

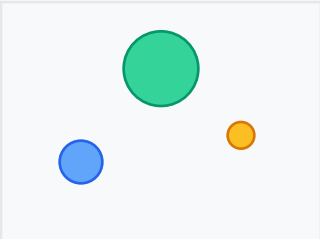
Syntax:
```
df.plot.area()
```



6. Gantt Chart

Purpose: Maps distinct structural phases, target production operations, and durations over calendar timelines.

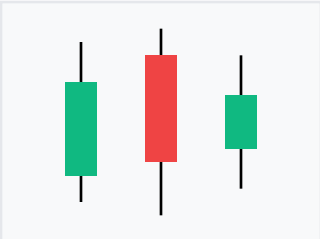
Syntax:
```
plt.barh(tasks, left=starts, width=durations)
```



7. Bubble Chart

Purpose: Enriches standard scatter coordinates by sizing markers proportionally to introduce a third parameter value scale.

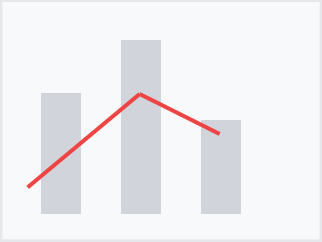
Syntax:
```
plt.scatter(x, y, s=sizes)
```



8. Candlestick Chart

Purpose: Tracks extreme opening, closing, high, and low pricing intervals in high-frequency trading applications.

Syntax:
```
go.Candlestick(open=o, high=h, low=l, close=c)
```

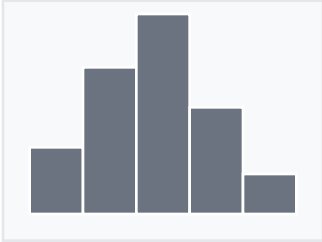


9. Dual-Axis Chart

Purpose: Plots distinct dataset values against two completely separate metric bounds across a shared timeline horizon.

Syntax:

```
ax2 = ax1.twinx()
```



10. Histogram

Purpose: Groups raw continuous dataset numbers into specific bins to highlight density frequency patterns.

Syntax:

```
plt.hist(data, bins=10)
```

11. Pie Chart

Purpose: Breaks down individual composition segments relative to the entire static uniform circular totality structure.

Syntax:

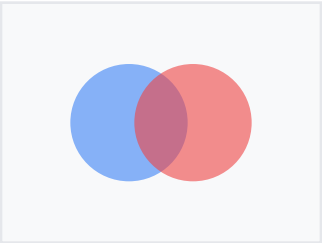
```
plt.pie(values,
labels=labels)
```

12. Donut Chart

Purpose: Modifies structural layout constraints of pie structures, placing categorical callouts in clean peripheral weights.

Syntax:

```
plt.pie(values);
circle=plt.Circle((0,0),
0.7, color='white');
ax.add_artist(circle)
```

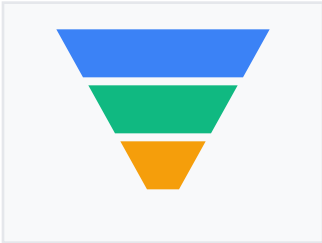


13. Venn Diagram

Purpose: Highlights subset intersections, logical commonalities, and mutual components across multiple entities.

Syntax:

```
venn2(subsets=(group1,
group2, common))
```



14. Funnel Chart

Purpose: Tracks specific progressive drop-off conversion layers down complex pipelines sequentially.

Syntax:

```
px.funnel(df, x='values',
y='stages')
```

15. Radar Chart

Purpose: Visualizes multivariable performance characteristics radially extending outwards from a unified origin point.

Syntax:

```
plt.subplot(111,
polar=True);
plt.plot(angles, values)
```

This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.